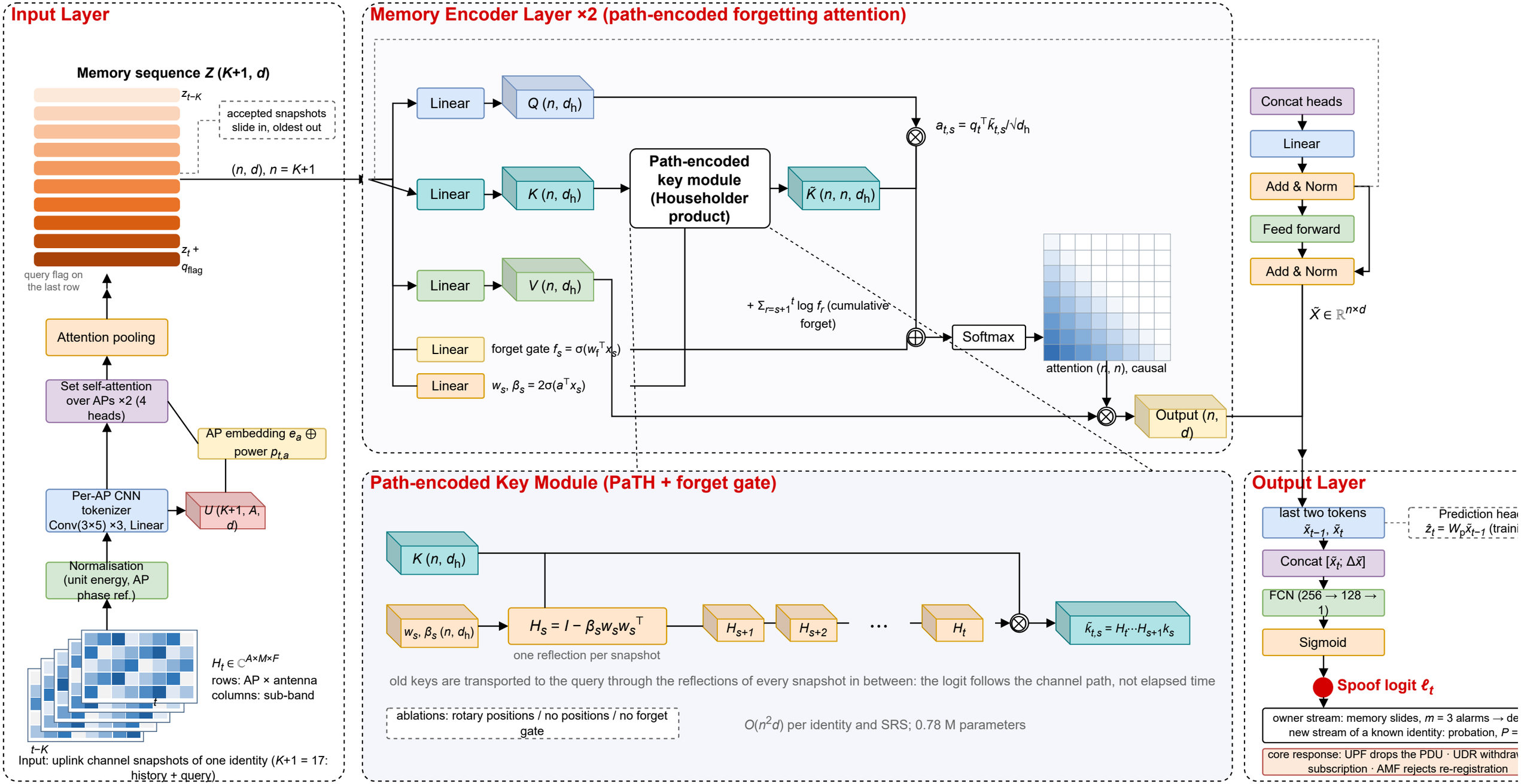
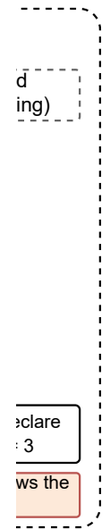

Input Layer
Memory sequence Z (K+1, d)
z_{t−K}
accepted snapshots slide in, oldest out
(n, d), n = K+1
z_t + q_flag
query flag on the last row
Attention pooling
Set self-attention over APs ×2 (4 heads)
AP embedding e_a ⊕ power p_{t,a}
Per-AP CNN tokenizer Conv(3×5) ×3, Linear
U (K+1, A, d)
Normalisation (unit energy, AP phase ref.)
H_t ∈ ℂ^{A×M×F}
rows: AP × antenna
columns: sub-band
t
t−K
Input: uplink channel snapshots of one identity (K+1 = 17: history + query)
Memory Encoder Layer ×2 (path-encoded forgetting attention)
Linear
Q (n, d_h)
K (n, d_h)
V (n, d_h)
Path-encoded key module (Householder product)
K̃ (n, n, d_h)
a_{t,s} = q_t^⊤ k̃_{t,s}/√d_h
+ Σ_{r=s+1}^{t} log f_r (cumulative forget)
Linear
forget gate f_s = σ(w_f^⊤ x_s)
Linear
w_s, β_s = 2σ(a^⊤ x_s)
Softmax
attention (n, n), causal
Output (n, d)
Concat heads
Linear
Add & Norm
Feed forward
Add & Norm
X̃ ∈ ℝ^{n×d}
Path-encoded Key Module (PaTH + forget gate)
K (n, d_h)
w_s, β_s (n, d_h)
H_s = I − β_s w_s w_s^⊤
H_{s+1}
H_{s+2}
H_t
k̃_{t,s} = H_t ··· H_{s+1} k_s
one reflection per snapshot
old keys are transported to the query through the reflections of every snapshot in between: the logit follows the channel path, not elapsed time
ablations: rotary positions / no positions / no forget gate
O(n²d) per identity and SRS; 0.78 M parameters
Output Layer
last two tokens x̃_{t−1}, x̃_t
Prediction hea
ẑ_t = W_p x̃_{t−1} (traini
Concat [x̃_t; Δx̃]
FCN (256 → 128 → 1)
Sigmoid
Spoof logit ℓ_t
owner stream: memory slides, m = 3 alarms → de
new stream of a known identity: probation, P =
core response: UPF drops the PDU · UDR withdra
subscription · AMF rejects re-registration


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